

Visualisation Examples Catalogue

Decision-support toolkit for visualising accelerometer-derived movement behaviour data

32 generated visualisation examples: 9 worked case-study examples and 23 simulated data examples.

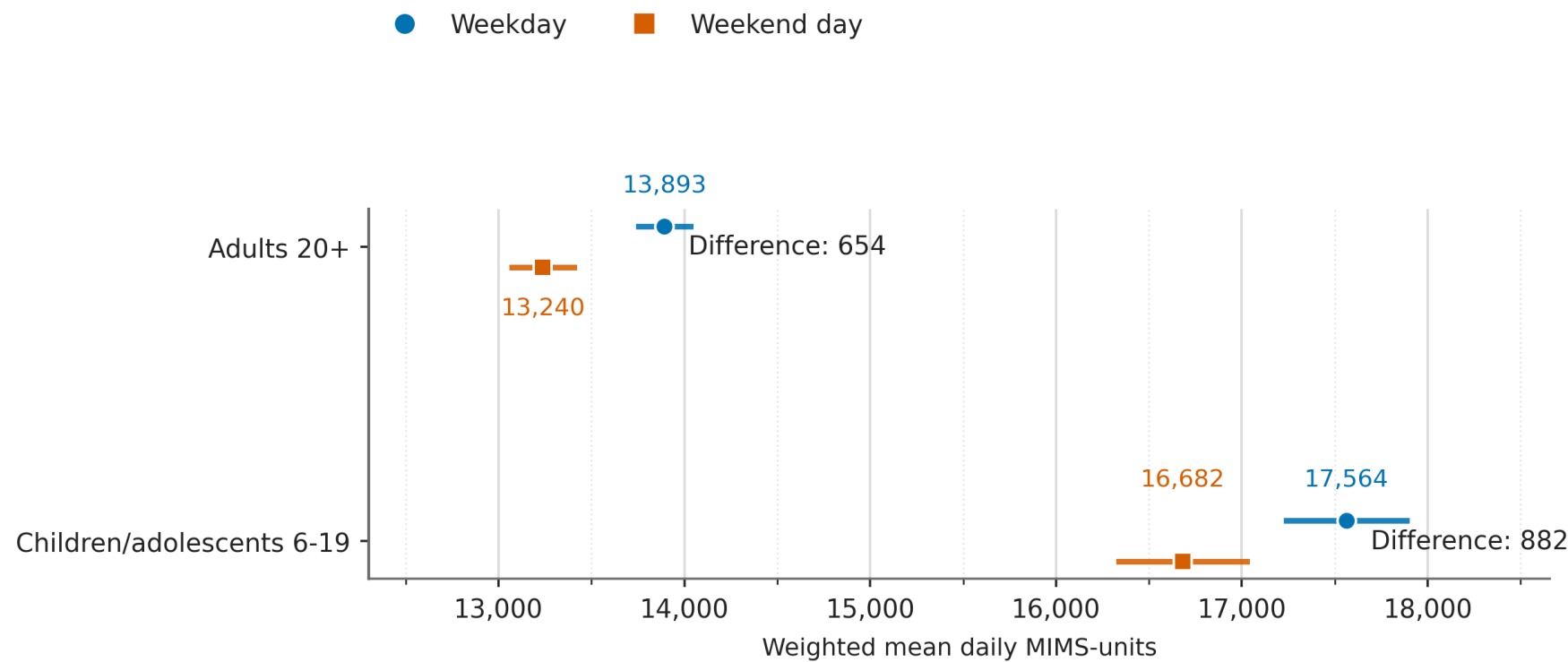
Worked case-study examples were generated from the reproduced NHANES 2011-2014 case-study outputs. Simulated data examples were generated from small mock datasets to illustrate recommendation-specific visual structures and should not be interpreted as evidence that the designs have been empirically tested.

Worked case study examples

Figures generated from the reproduced NHANES case-study dataset and associated summary outputs.

Worked example 1. Weekday and weekend-day physical activity by sample

Weekday and weekend-day physical activity by sample



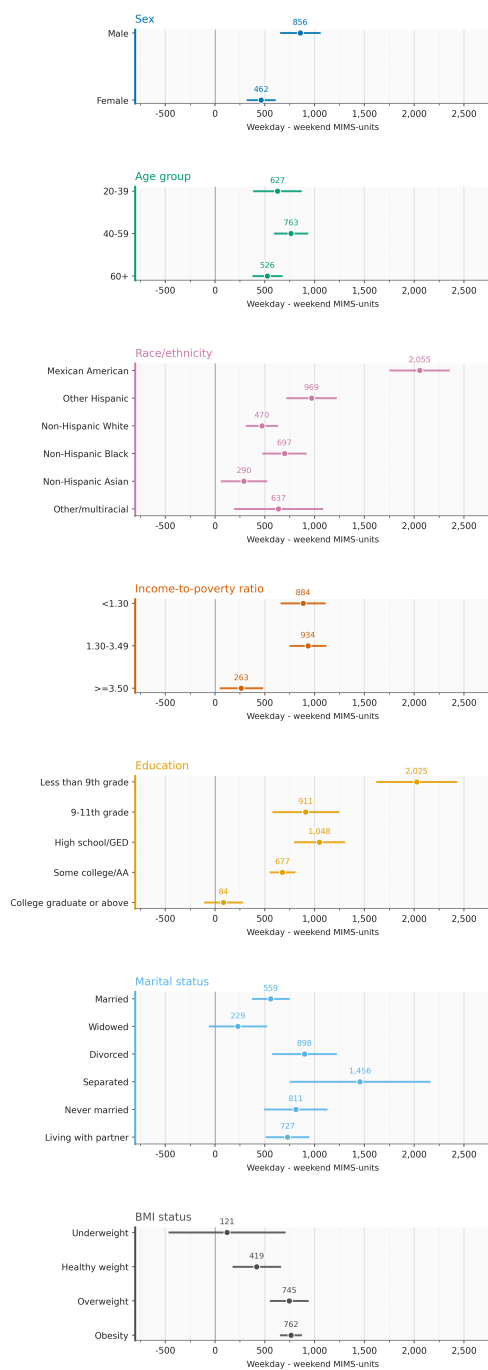
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. The x-axis is zoomed to make the paired day-type estimates readable and starts at 12,302 MIMS-units, not zero; values are shown in the original metric units.

Weighted mean daily MIMS-units on weekdays and weekend days for adults and children/adolescents. Points show estimates, horizontal intervals show 95% confidence intervals, and text labels report the descriptive weekday-minus-weekend difference. No statistical significance marker is shown.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 2. Weekday-weekend physical activity difference by subgroup: Adults 20+

Weekday-weekend physical activity difference by subgroup: Adults 20+



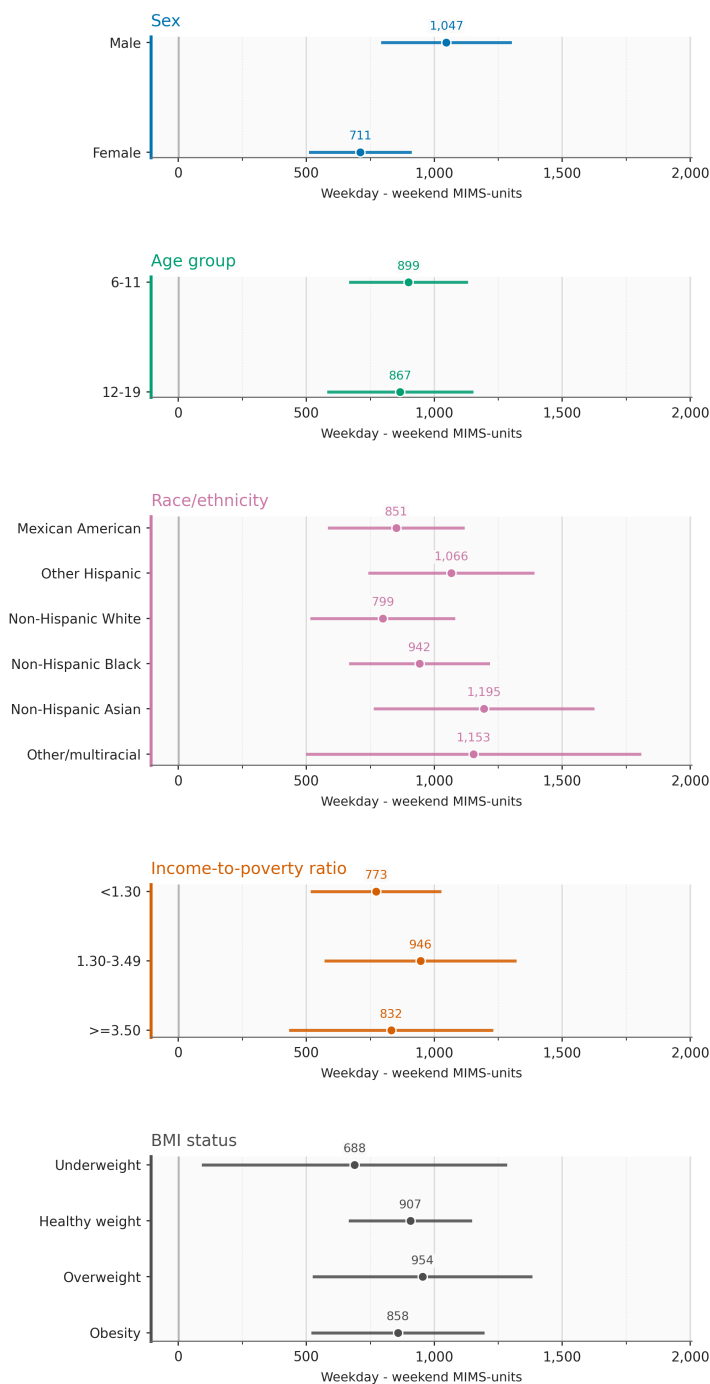
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMIS = Monitor-Independent Movement Summary units. Positive values indicate higher weekday activity. Panel colours identify subgroup domains.

Weighted weekday-minus-weekend MIMIS-unit differences by subgroup for Adults 20+. Positive values indicate higher weekday physical activity. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 3. Weekday-weekend physical activity difference by subgroup: Children/adolescents 6-19

Weekday-weekend physical activity difference by subgroup: Children/adolescents 6-19



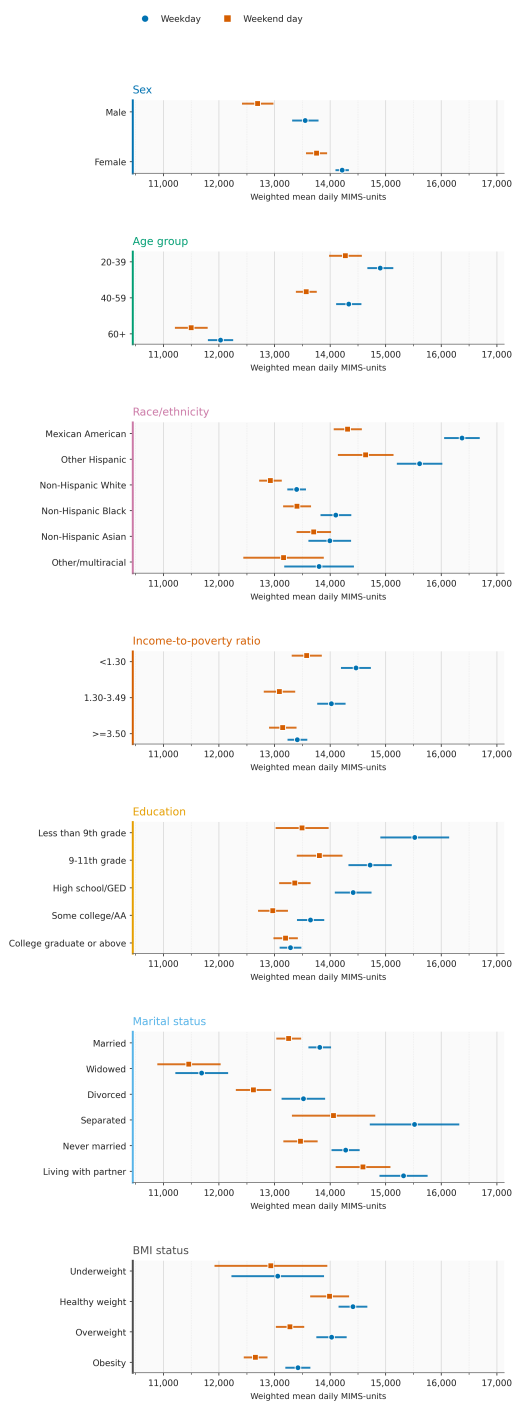
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. Positive values indicate higher weekday activity. Panel colours identify subgroup domains.

Weighted weekday-minus-weekend MIMS-unit differences by subgroup for Children/adolescents 6-19. Positive values indicate higher weekday physical activity. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 4. Weekday and weekend-day physical activity by subgroup: Adults 20+

Weekday and weekend-day physical activity by subgroup: Adults 20+



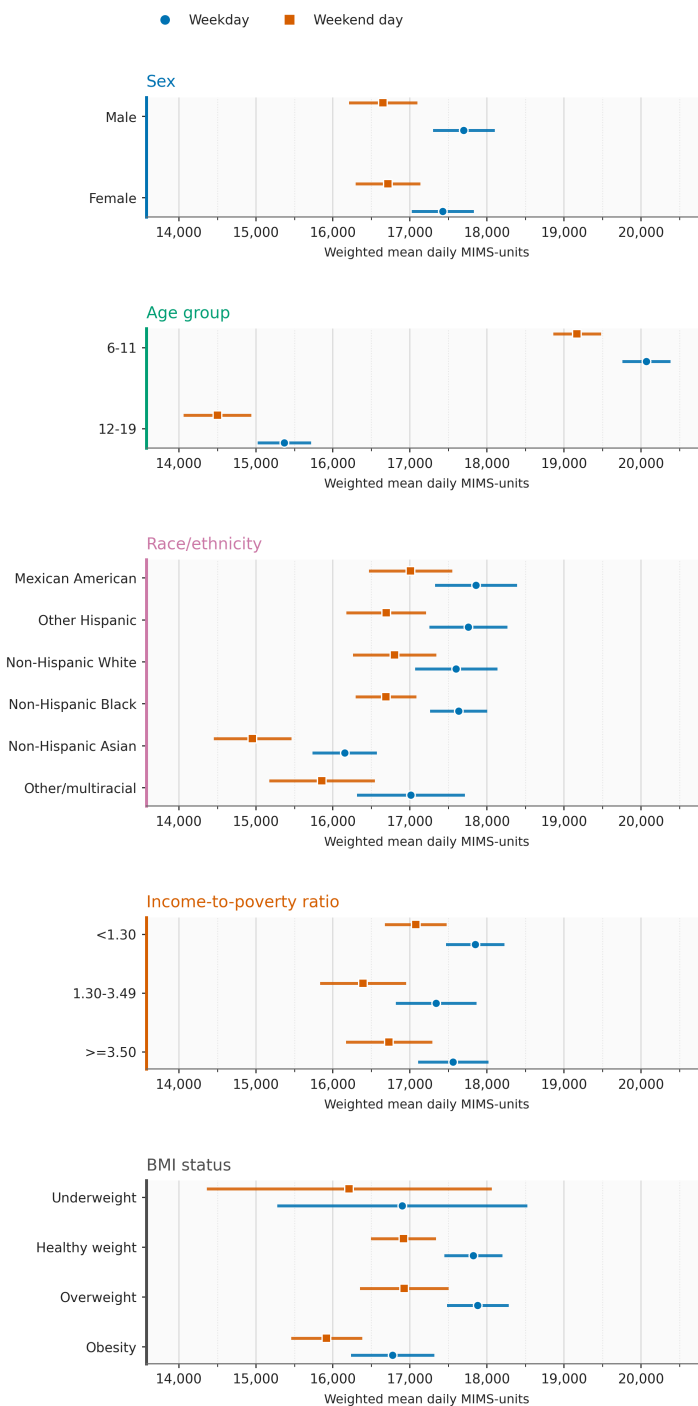
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. The x-axis is scaled to show subgroup comparisons; dotted minor gridlines help locate values. Panel colours identify subgroup domains.

Weighted mean daily MIMS-units on weekdays and weekend days by subgroup for Adults 20+. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 5. Weekday and weekend-day physical activity by subgroup: Children/adolescents 6-19

Weekday and weekend-day physical activity by subgroup: Children/adolescents 6-19



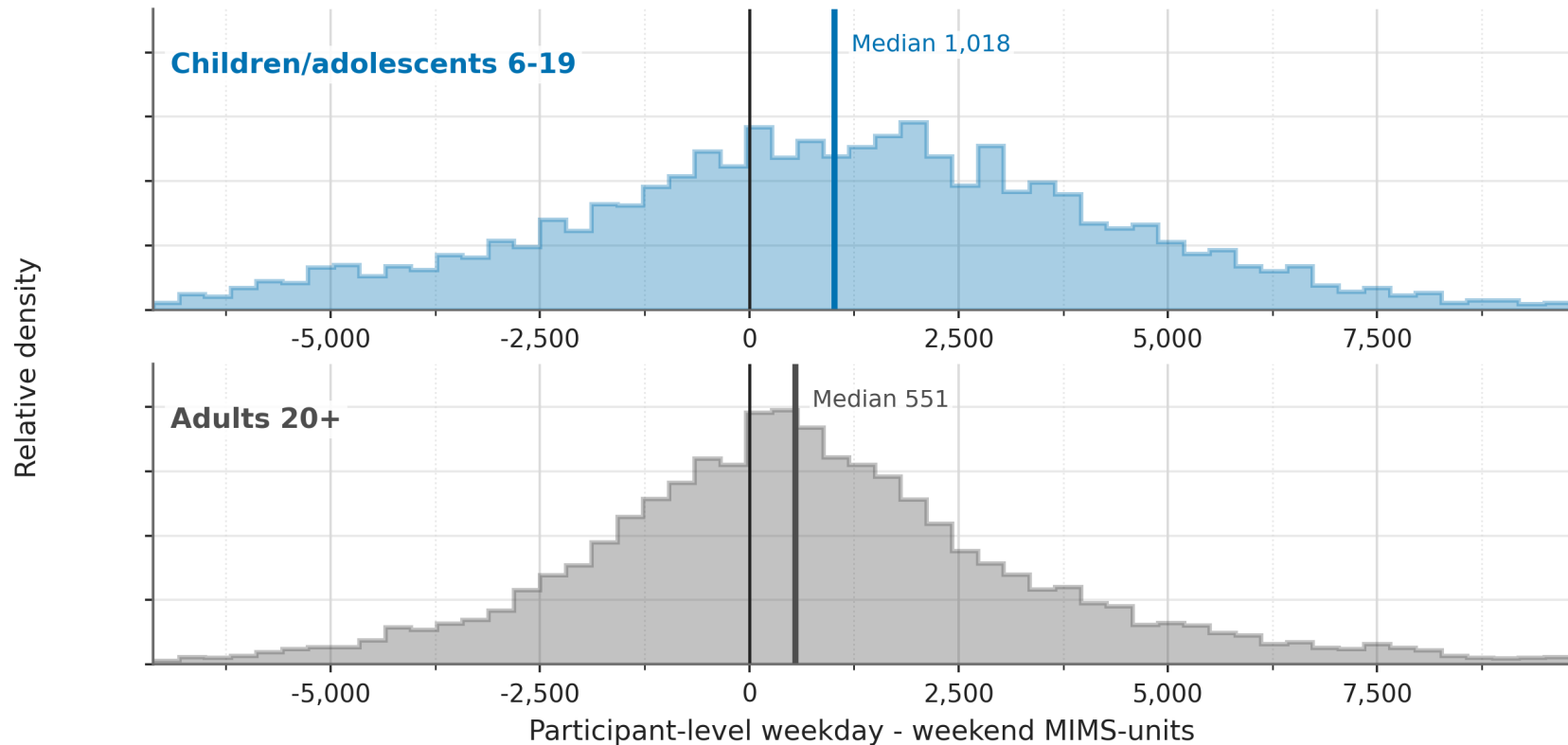
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. The x-axis is scaled to show subgroup comparisons; dotted minor gridlines help locate values. Panel colours identify subgroup domains.

Weighted mean daily MIMS-units on weekdays and weekend days by subgroup for Children/adolescents 6-19. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 6. Distribution of individual weekday-weekend differences

Distribution of individual weekday-weekend differences



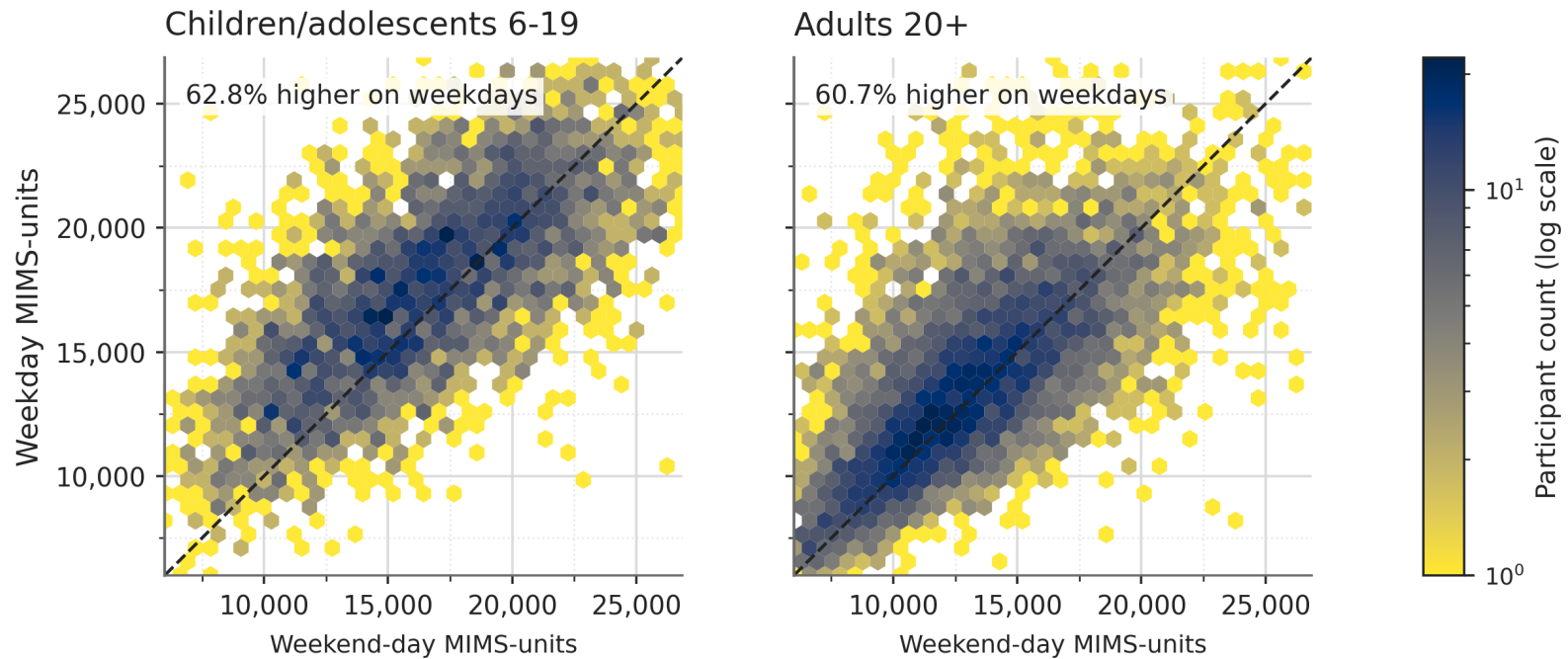
Distributions use unweighted participant-level differences and show the central 98% of values to keep the distribution readable; median lines are calculated from all retained participants. Positive values indicate higher weekday activity.

Distribution of participant-level weekday-minus-weekend MIMS-unit differences for adults and children/adolescents. Positive values indicate higher weekday physical activity. The axis shows the central 98% of values for readability.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 7. Participant-level relationship between weekday and weekend-day activity

Participant-level relationship between weekday and weekend-day activity



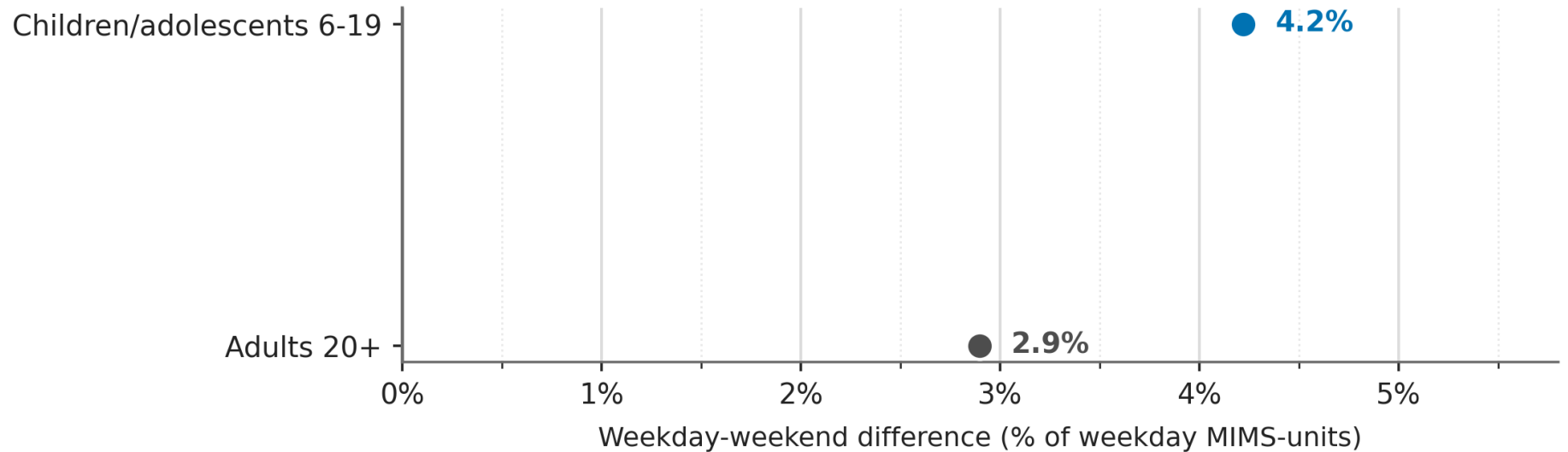
Each hexagon represents participants with similar weekday and weekend-day MIMS values; darker colour means more participants. The dashed line marks equal weekday and weekend-day activity. Axes show the central 98% of participant values for readability.

Hexbin plots showing the participant-level relationship between mean weekend-day and weekday MIMS-units. The dashed line represents equal weekday and weekend-day activity; points above the line indicate higher weekday activity.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 8. Percentage weekday-weekend difference by sample

Percentage weekday-weekend difference by sample



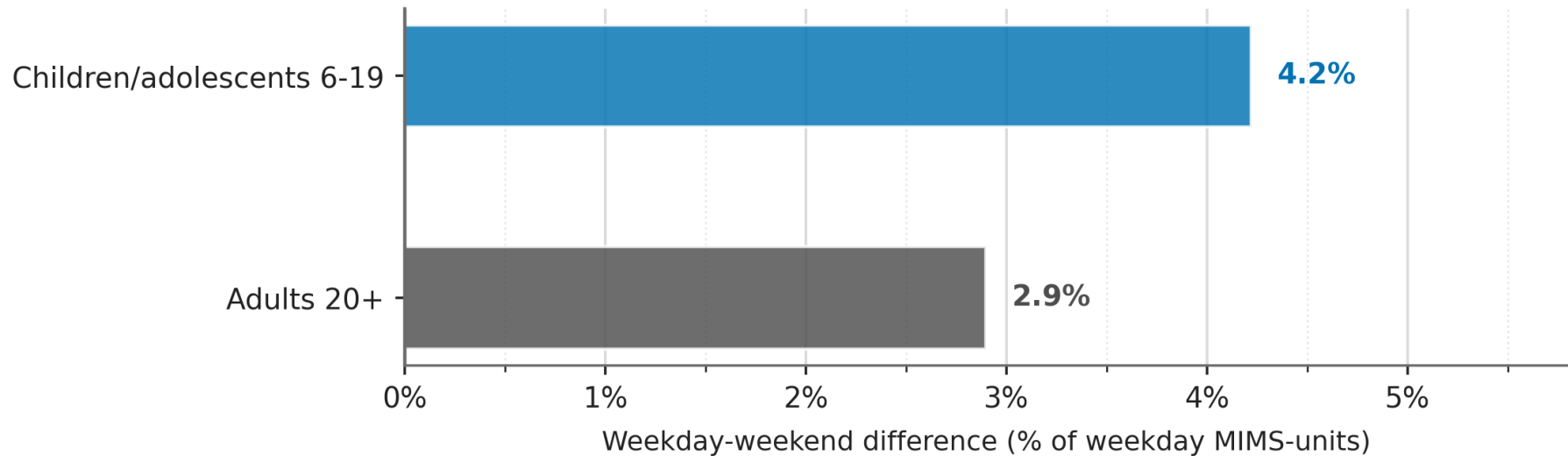
Dots show reconstructed weighted percentage differences. Intervals are not drawn in this example because the percentage column is treated as secondary descriptive context; use a point-range plot when uncertainty is central to the message.

Reconstructed weekday-weekend percentage differences relative to weekday MIMS-units for adults and children/adolescents. Dots show weighted summary percentages without intervals.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Worked example 9. Conditional bar-chart example: percentage difference by sample

Conditional bar-chart example: percentage difference by sample



This bar version is a conditional alternative because the percentage difference has a meaningful zero and the message is magnitude from zero. Prefer point-range or dot plots when uncertainty, distribution, or precise comparisons are the main message.

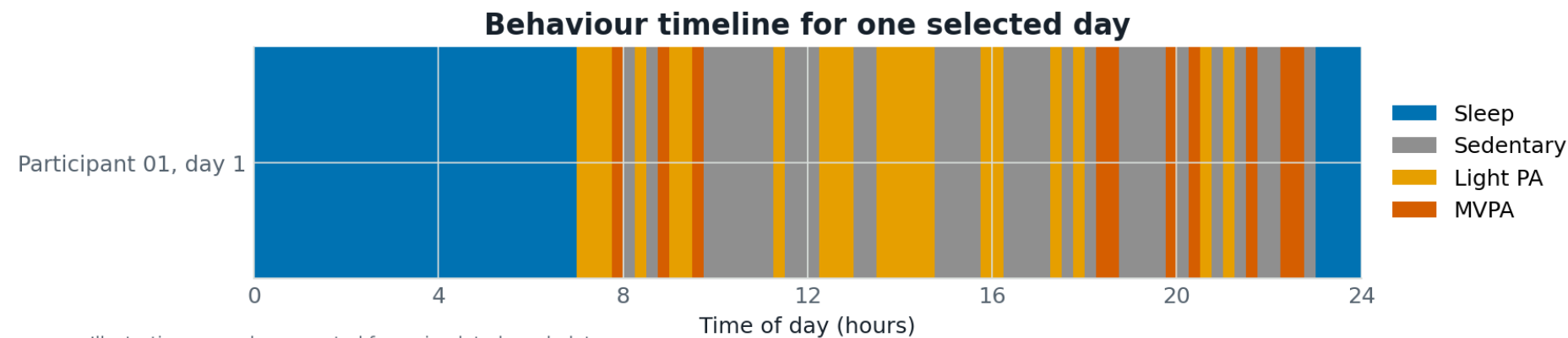
Conditional bar-chart version of the reconstructed weekday-weekend percentage differences relative to weekday MIMS-units. Bar length represents percentage difference from zero.

Source: Worked case study, reproduced NHANES 2011-2014 data.

Simulated data examples

Figures generated from small simulated mock datasets to illustrate recommendation-specific visual structures.

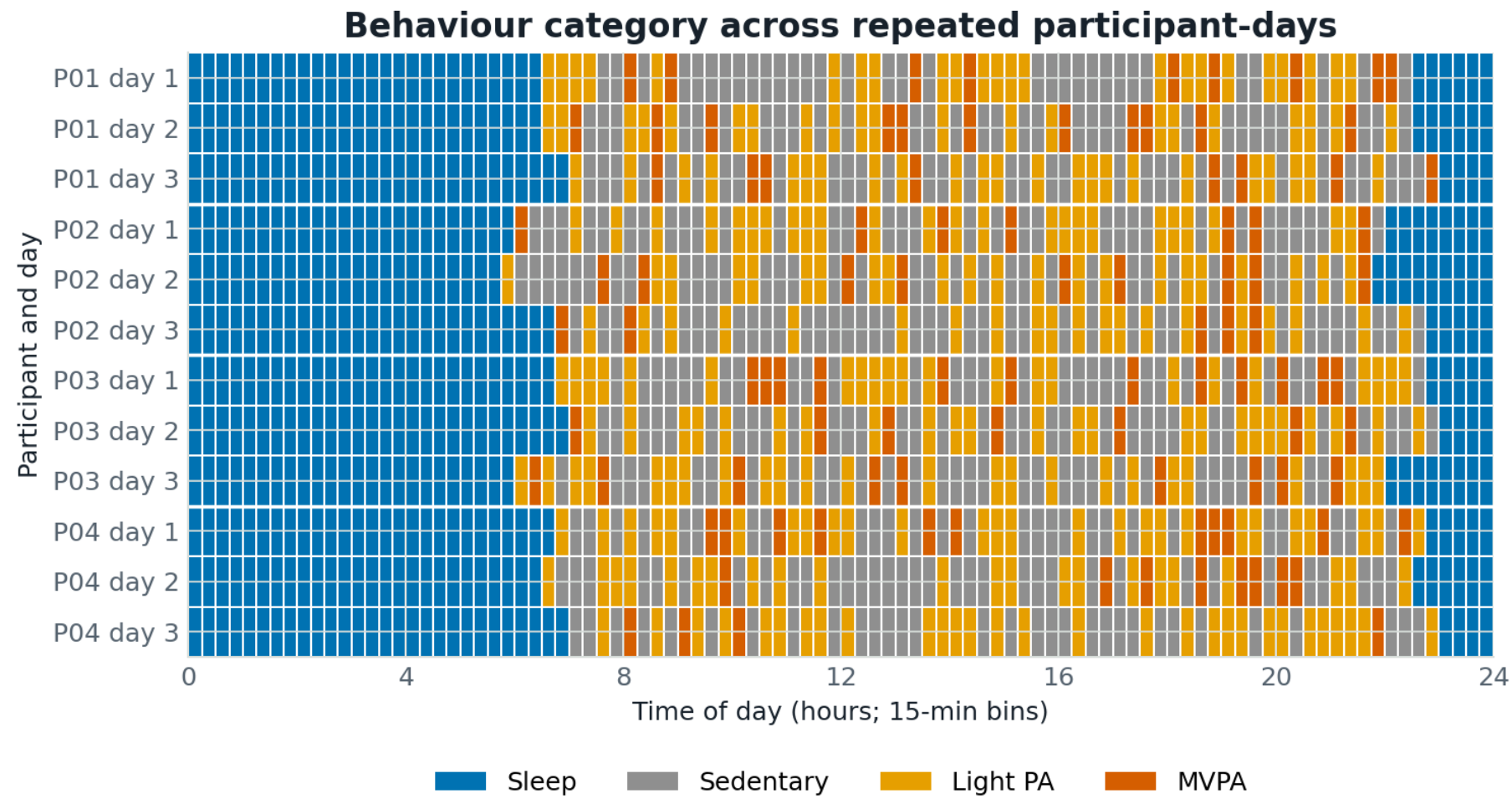
Simulated example 1. Behaviour timeline (tile plot)



Simulated data example illustrating the visual structure for the recommendation: Behaviour timeline (tile plot).

Source: Simulated mock data.

Simulated example 2. Behaviour-by-time heatmap

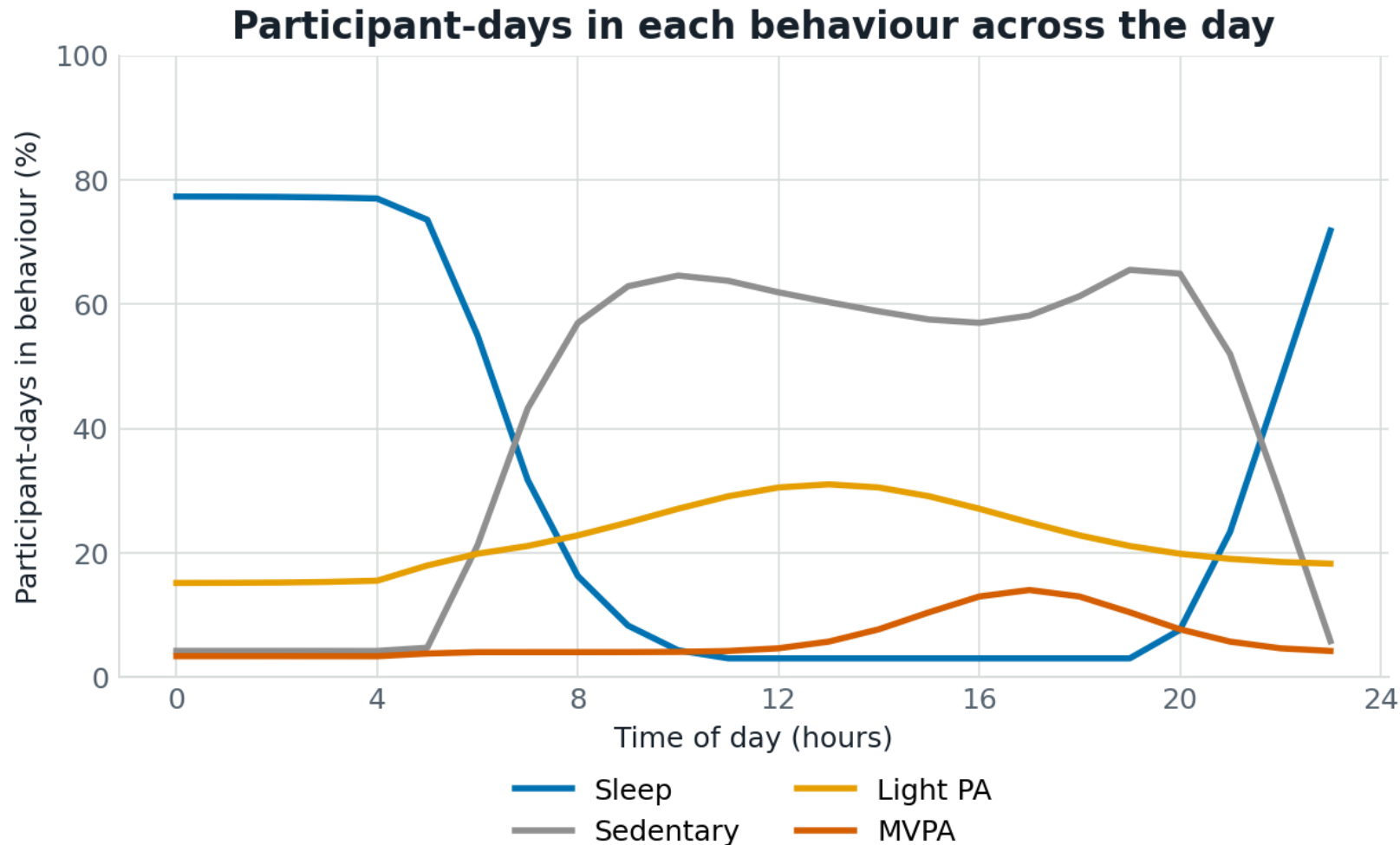


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Behaviour-by-time heatmap.

Source: Simulated mock data.

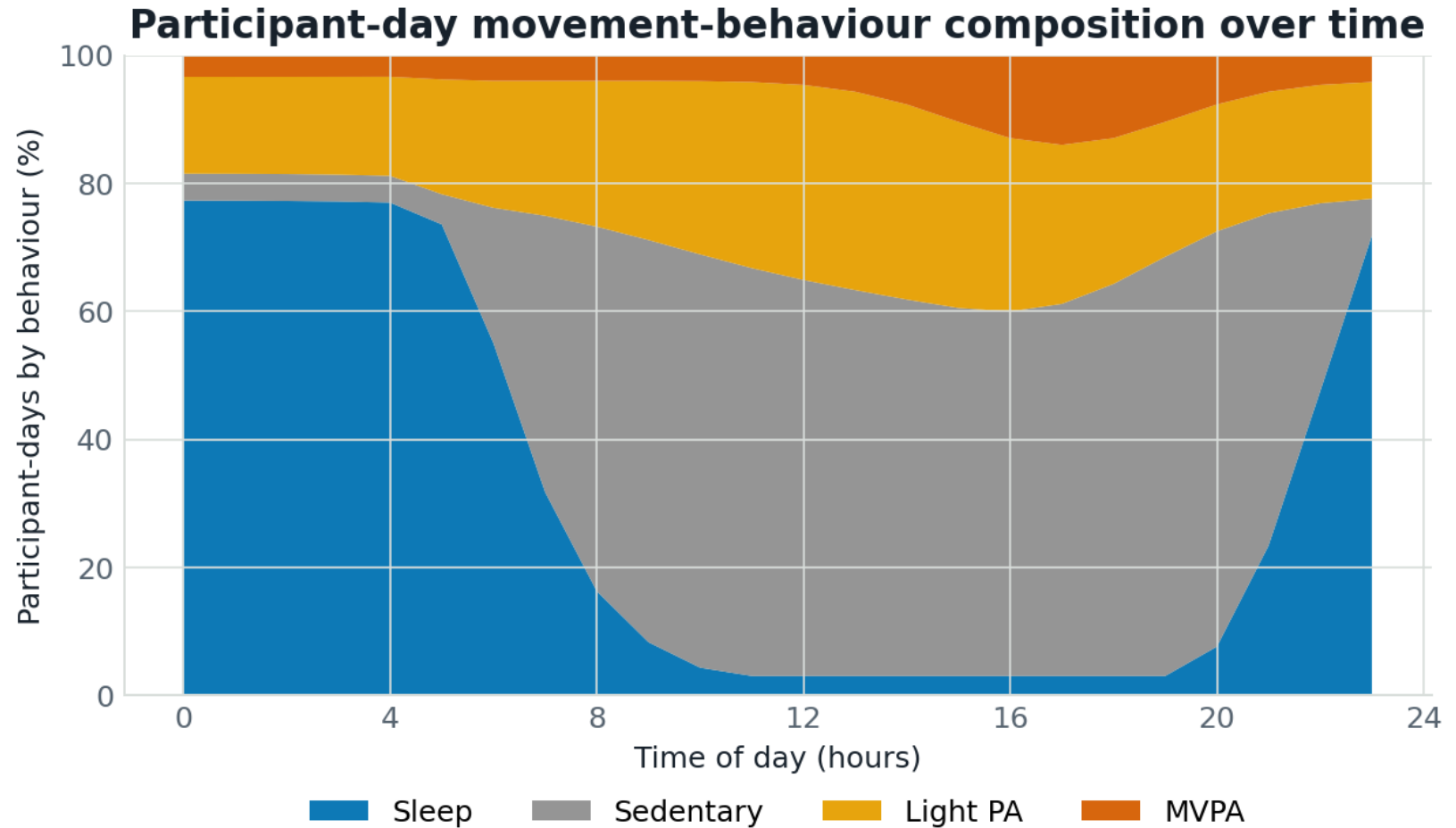
Simulated example 3. Proportion-over-time profile



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Proportion-over-time profile.
Source: Simulated mock data.

Simulated example 4. Stacked area profile

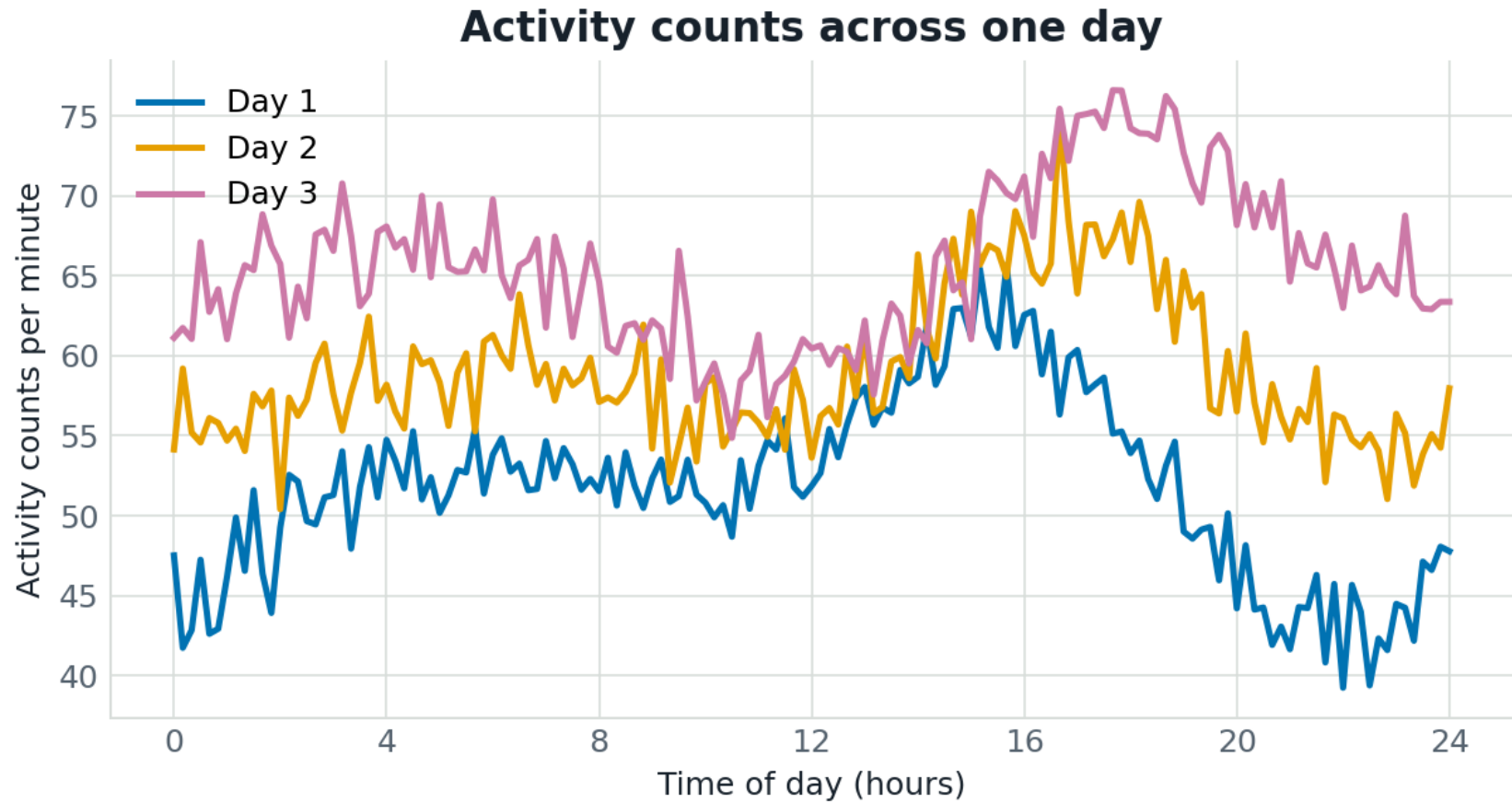


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Stacked area profile.

Source: Simulated mock data.

Simulated example 5. Time-series line plot

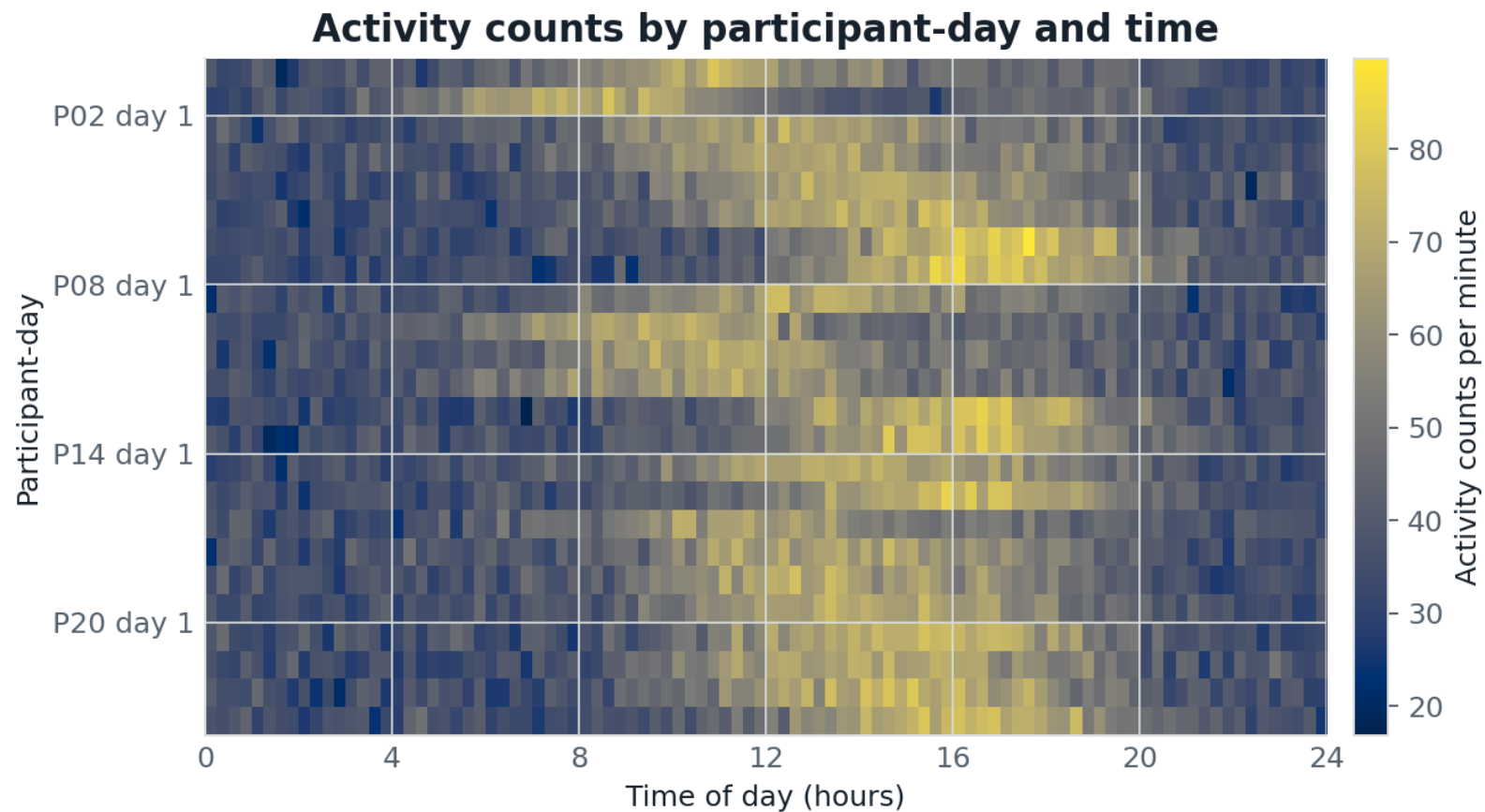


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Time-series line plot.

Source: Simulated mock data.

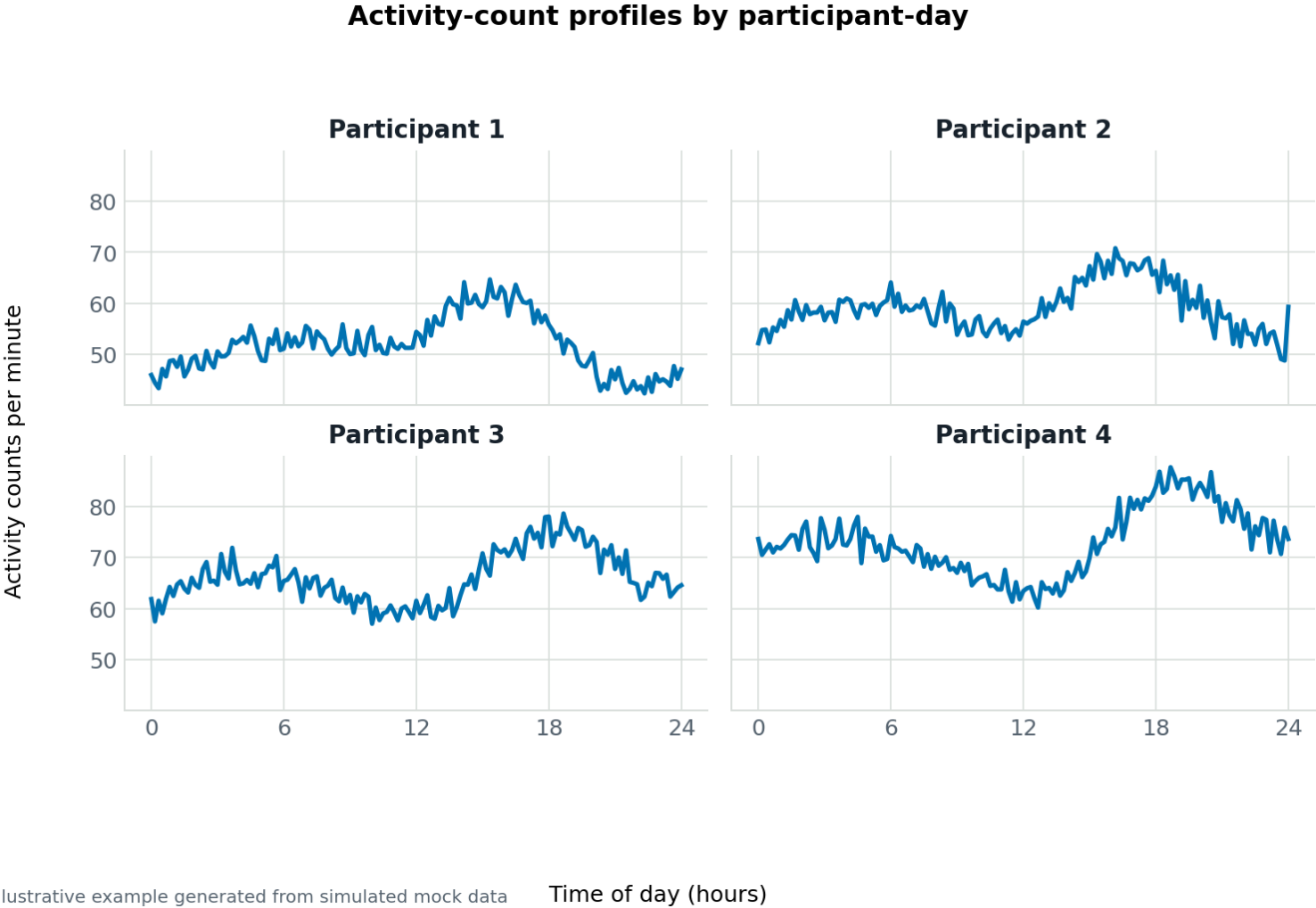
Simulated example 6. Observation-by-time heatmap



Illustrative example generated from simulated mock data

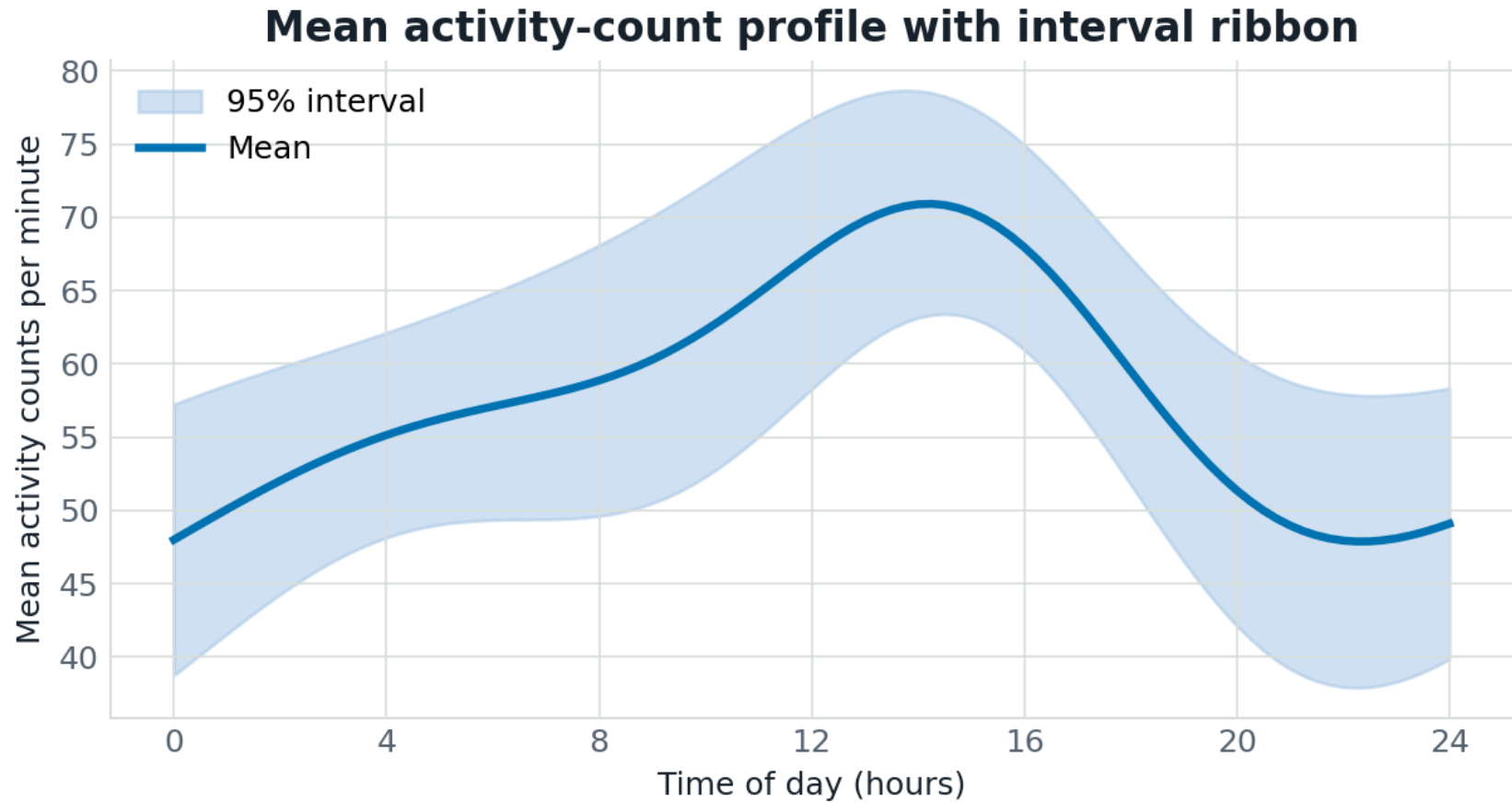
Simulated data example illustrating the visual structure for the recommendation: Observation-by-time heatmap.
Source: Simulated mock data.

Simulated example 7. Small-multiple time-series plots



Simulated data example illustrating the visual structure for the recommendation: Small-multiple time-series plots.
Source: Simulated mock data.

Simulated example 8. Summary time profile with interval ribbon

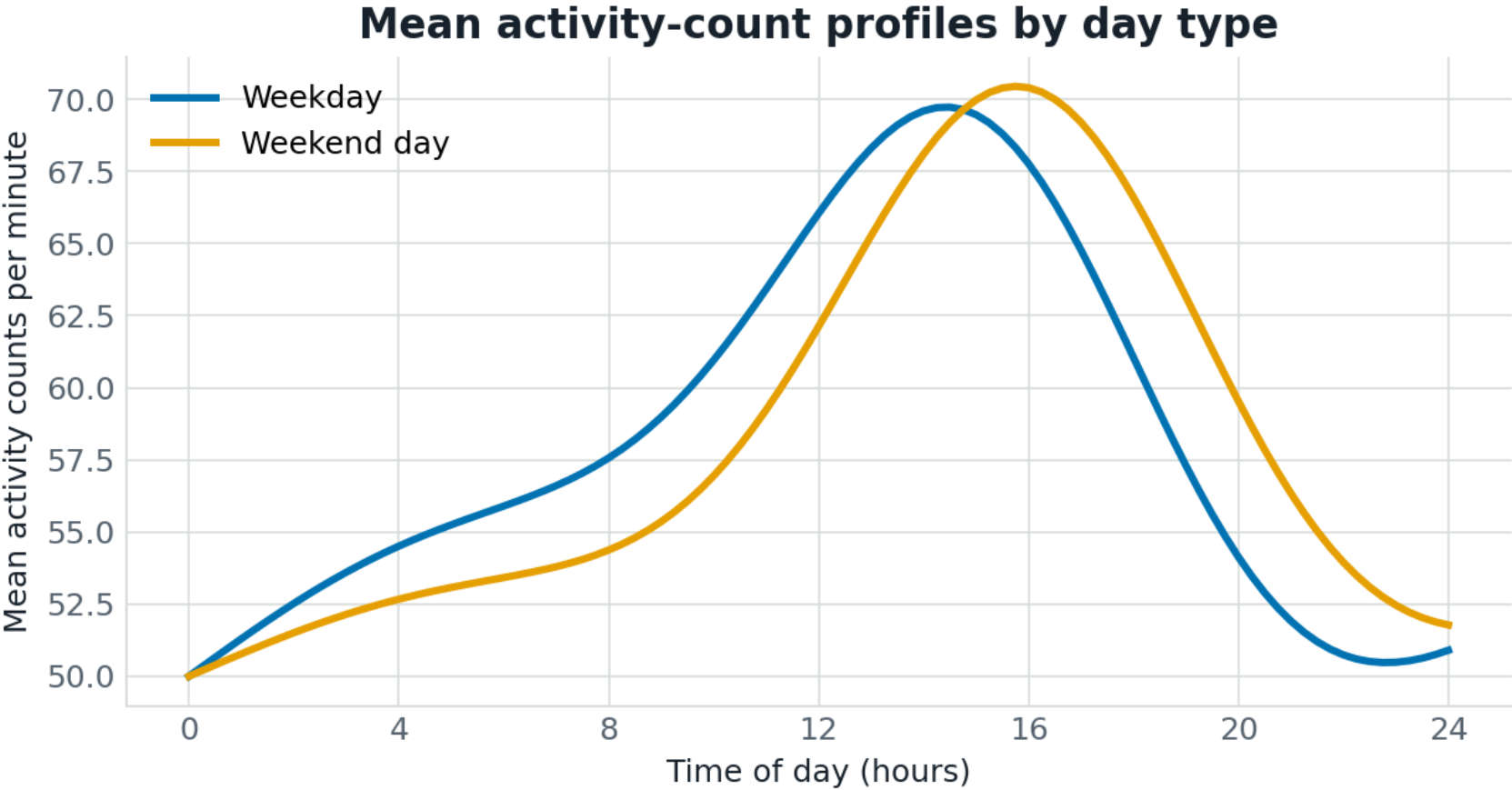


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Summary time profile with interval ribbon.

Source: Simulated mock data.

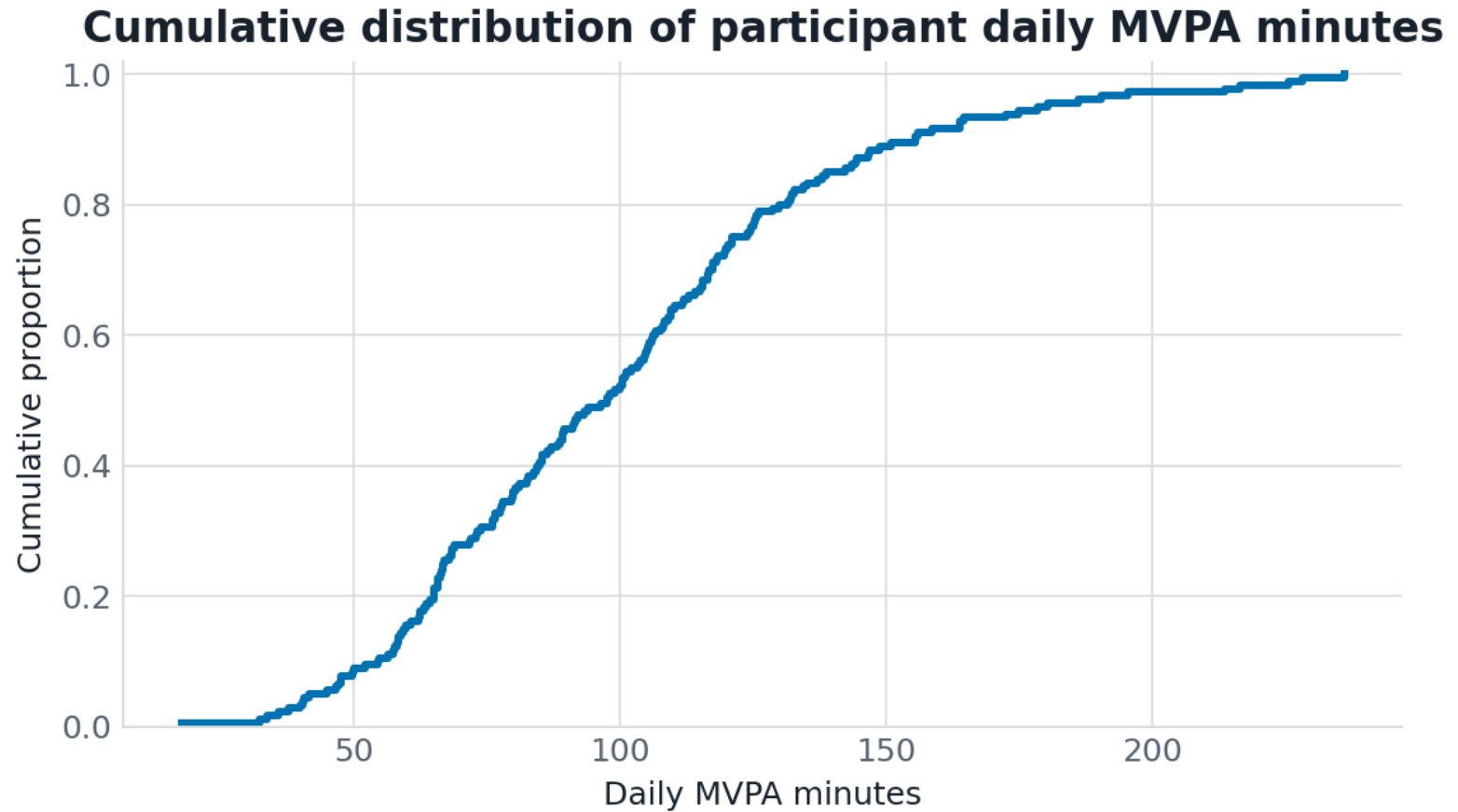
Simulated example 9. Summary time profile



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Summary time profile.
Source: Simulated mock data.

Simulated example 10. Empirical cumulative distribution (ECDF)

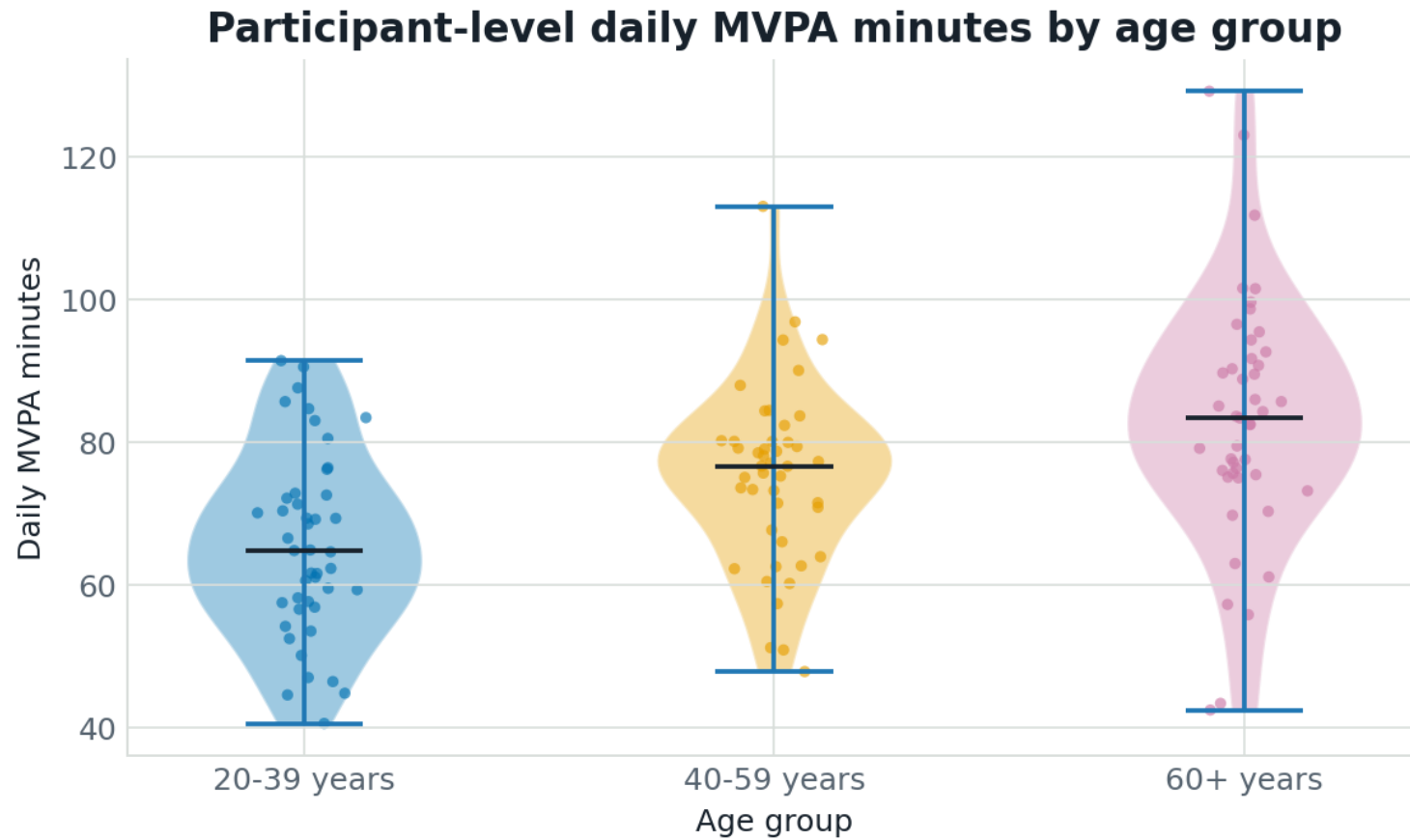


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Empirical cumulative distribution (ECDF).

Source: Simulated mock data.

Simulated example 11. Box or violin plot with raw points



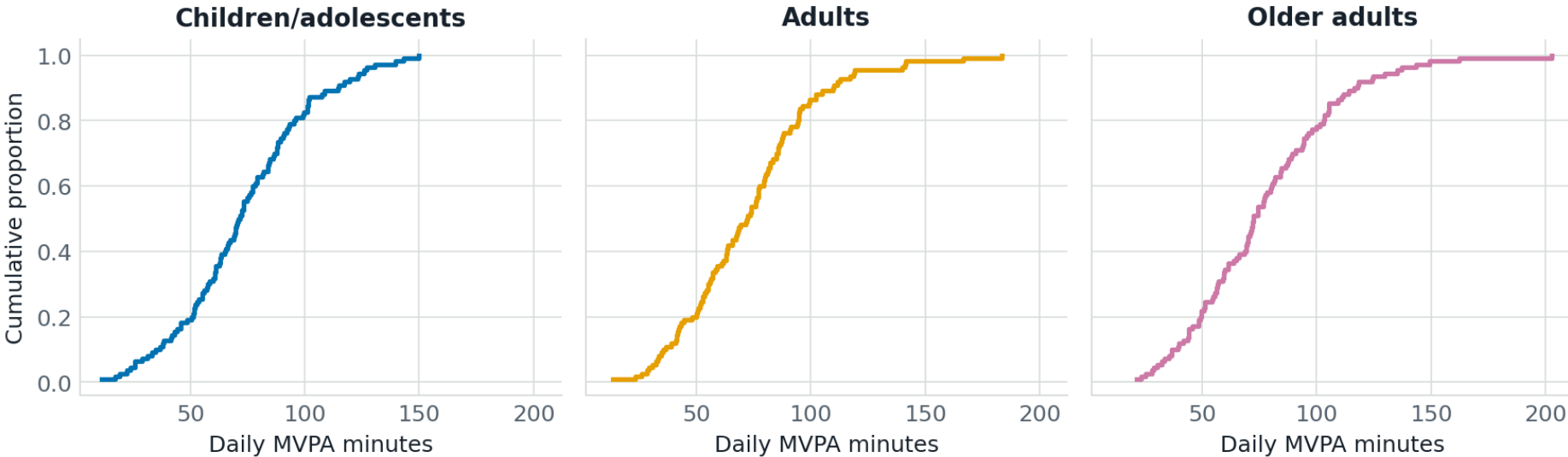
Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Box or violin plot with raw points.

Source: Simulated mock data.

Simulated example 12. Faceted density or ECDF plot

Daily MVPA minutes by sample group

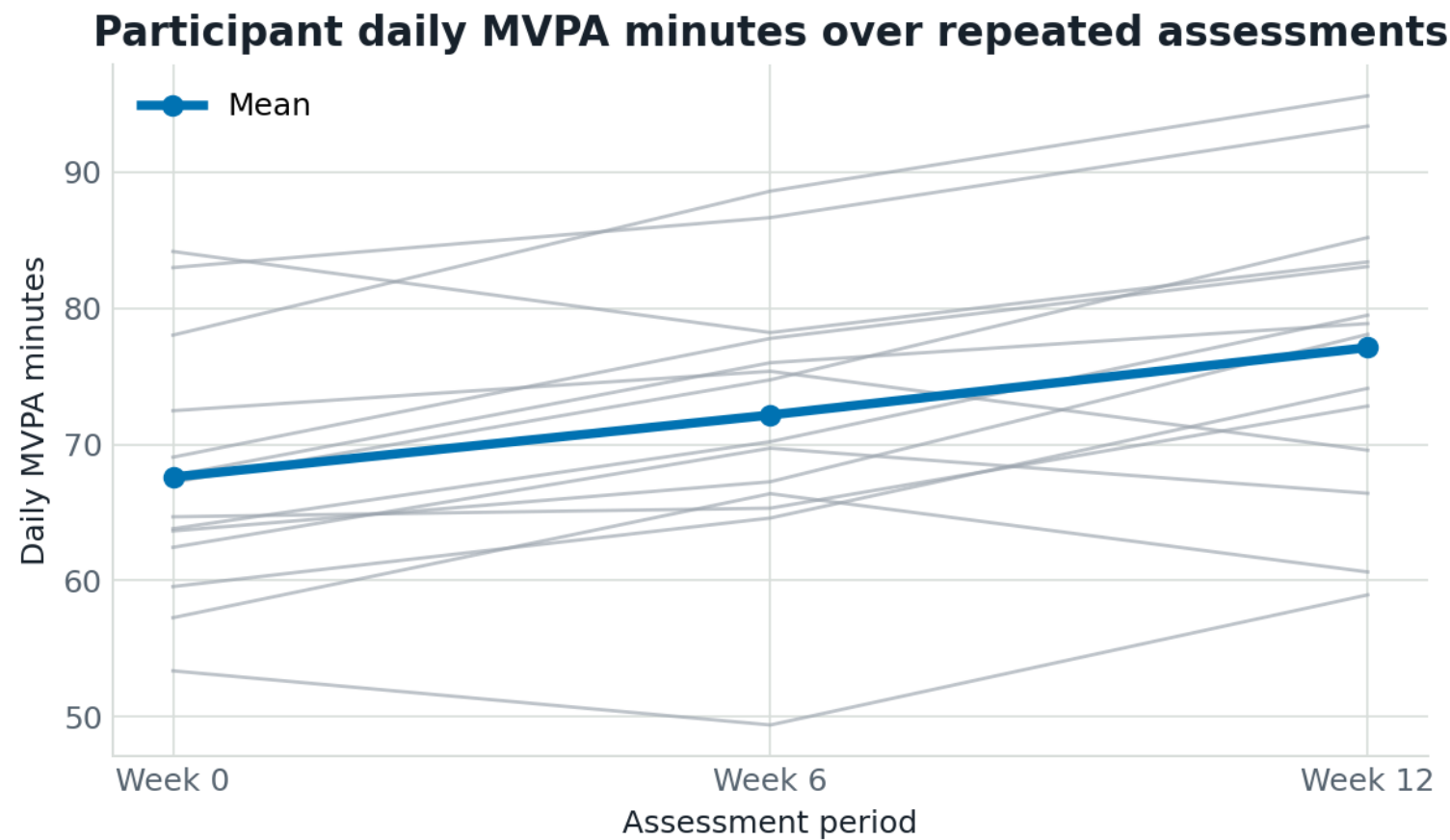


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Faceted density or ECDF plot.

Source: *Simulated mock data.*

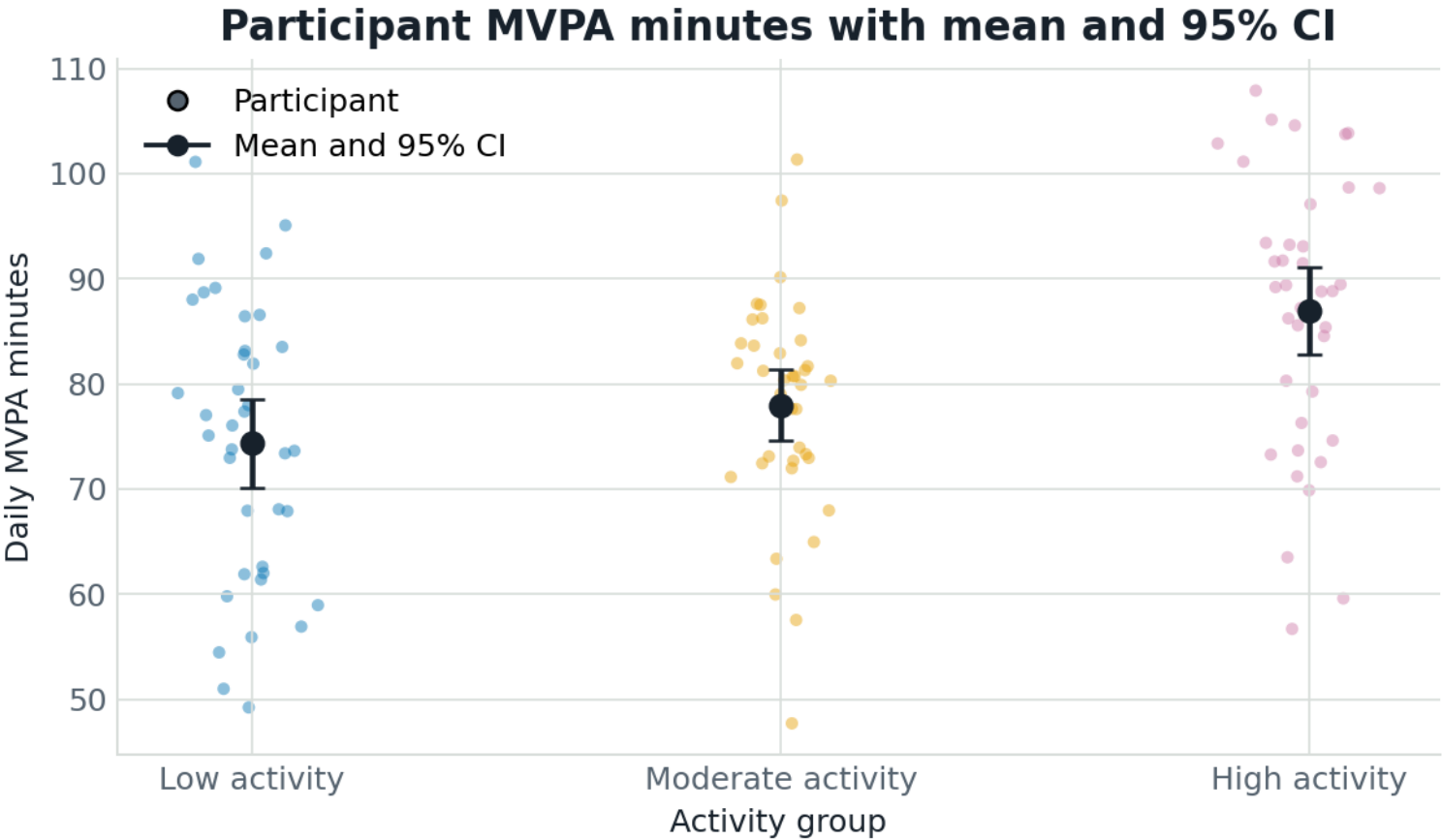
Simulated example 13. Repeated-measures line plot



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Repeated-measures line plot.
Source: Simulated mock data.

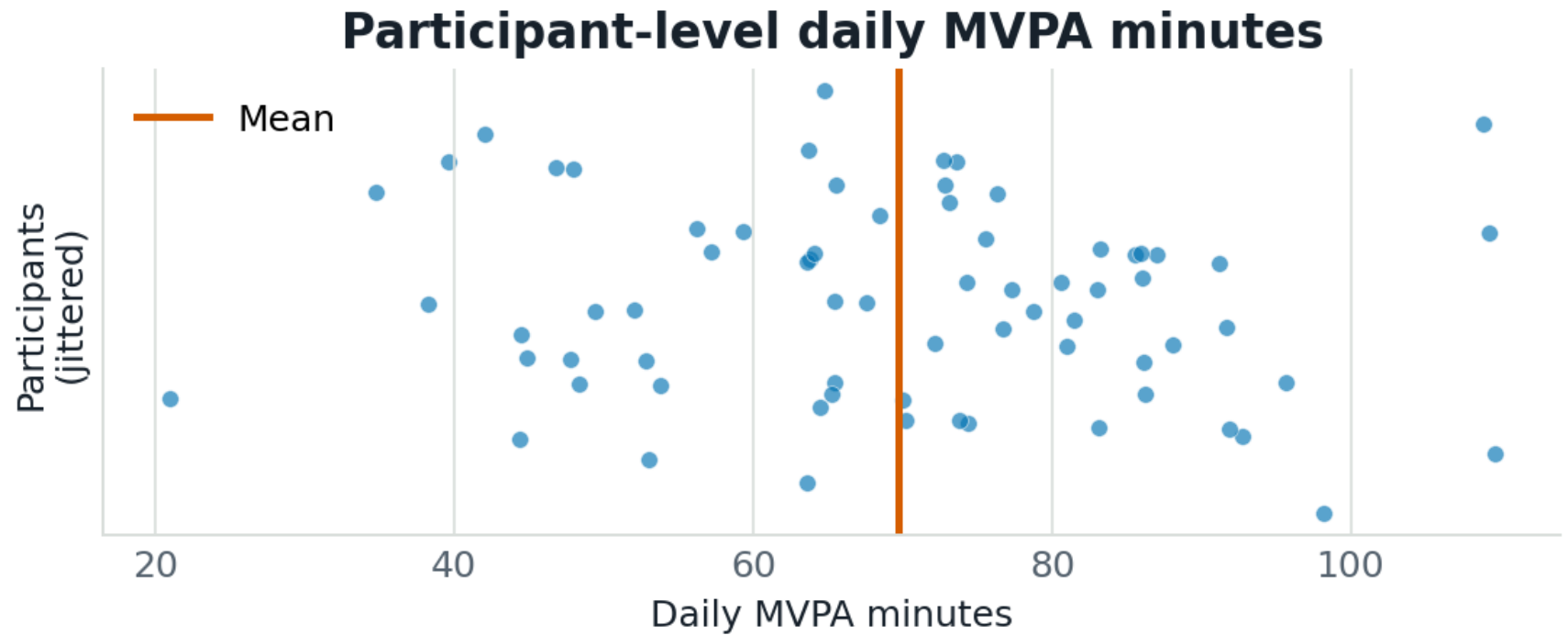
Simulated example 14. Dot plot with summary and interval



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Dot plot with summary and interval.
Source: Simulated mock data.

Simulated example 15. Dot plot of observed values



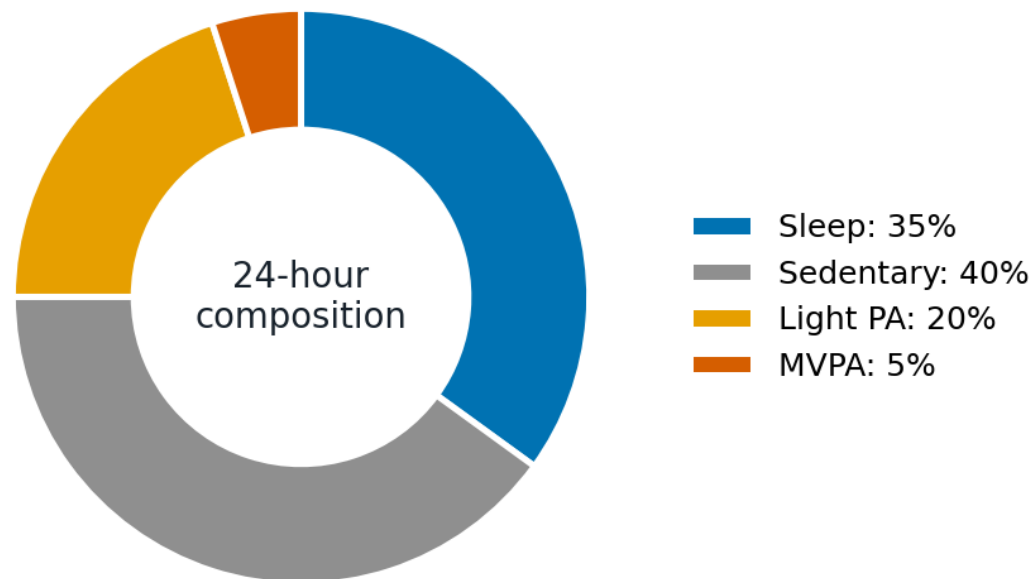
Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Dot plot of observed values.

Source: Simulated mock data.

Simulated example 16. Pie or doughnut chart

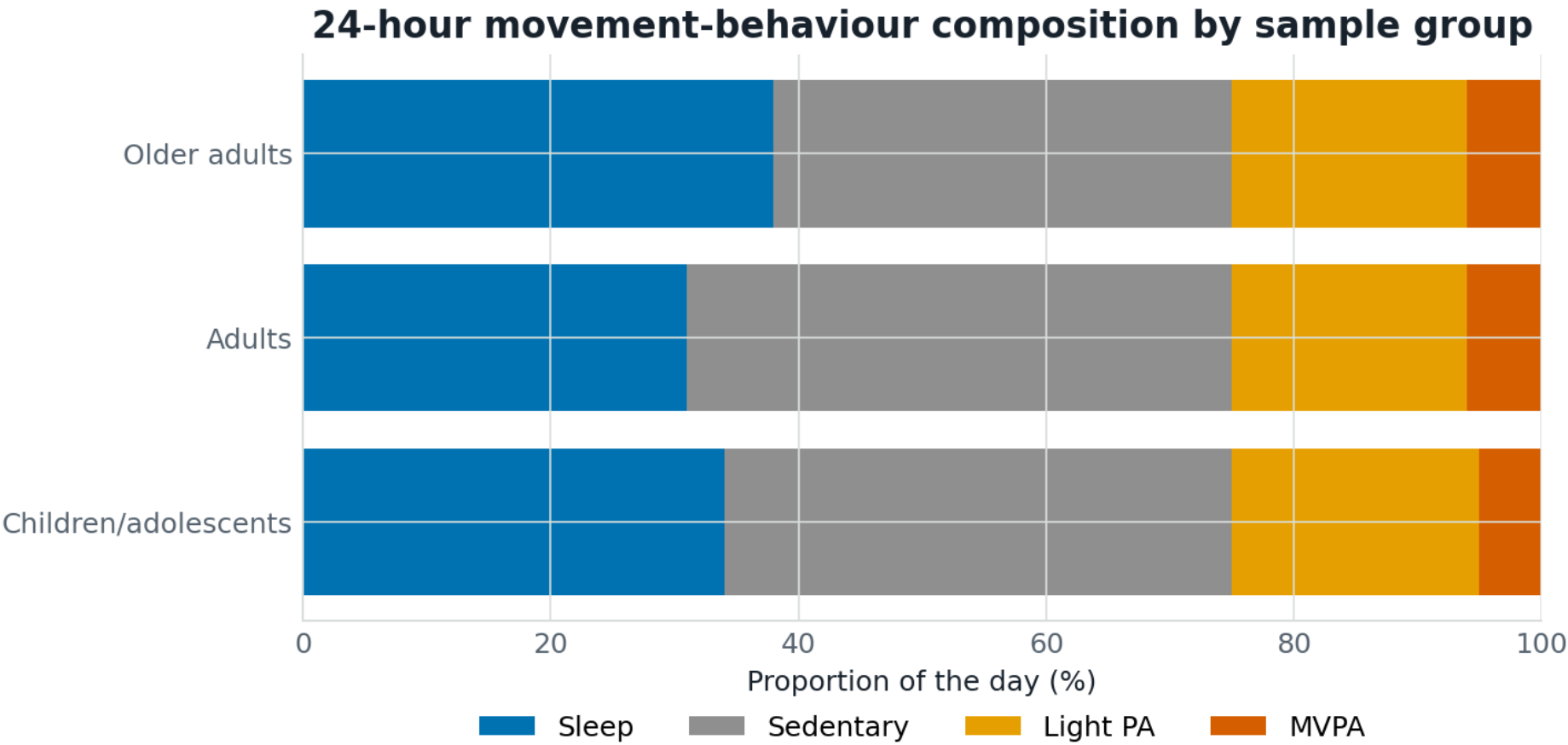
Example 24-hour movement-behaviour composition



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Pie or doughnut chart.
Source: Simulated mock data.

Simulated example 17. 100% stacked bar chart

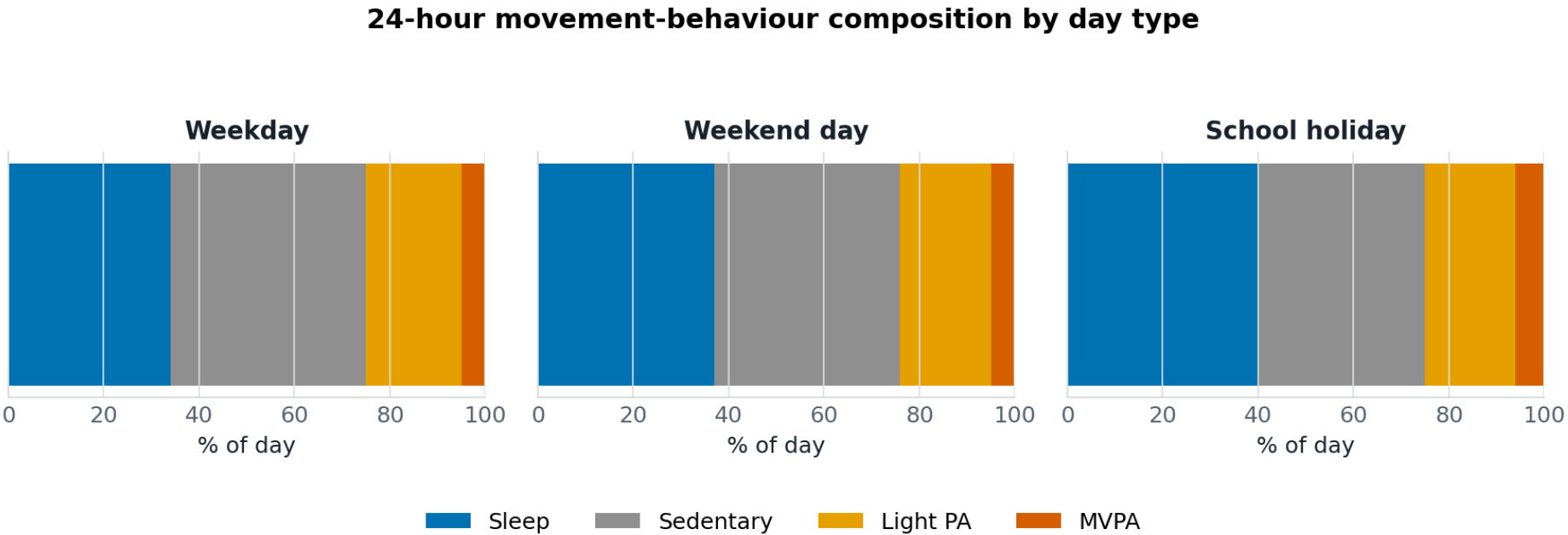


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: 100% stacked bar chart.

Source: Simulated mock data.

Simulated example 18. Small-multiple composition bars



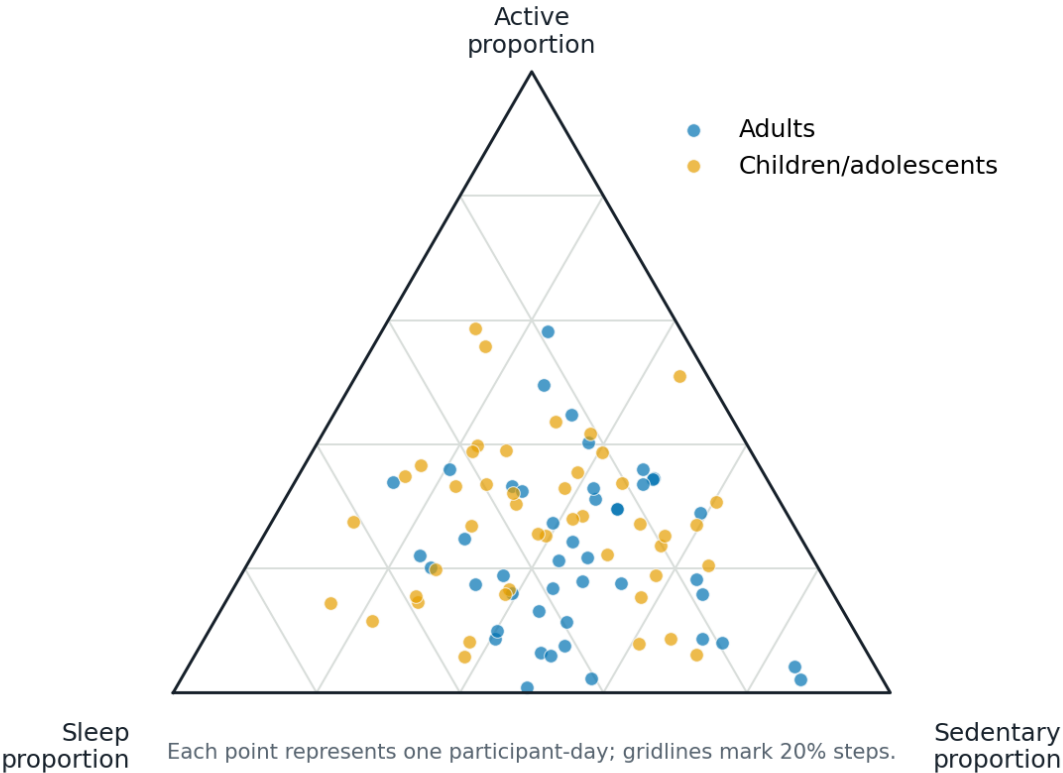
Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Small-multiple composition bars.

Source: Simulated mock data.

Simulated example 19. Ternary plot

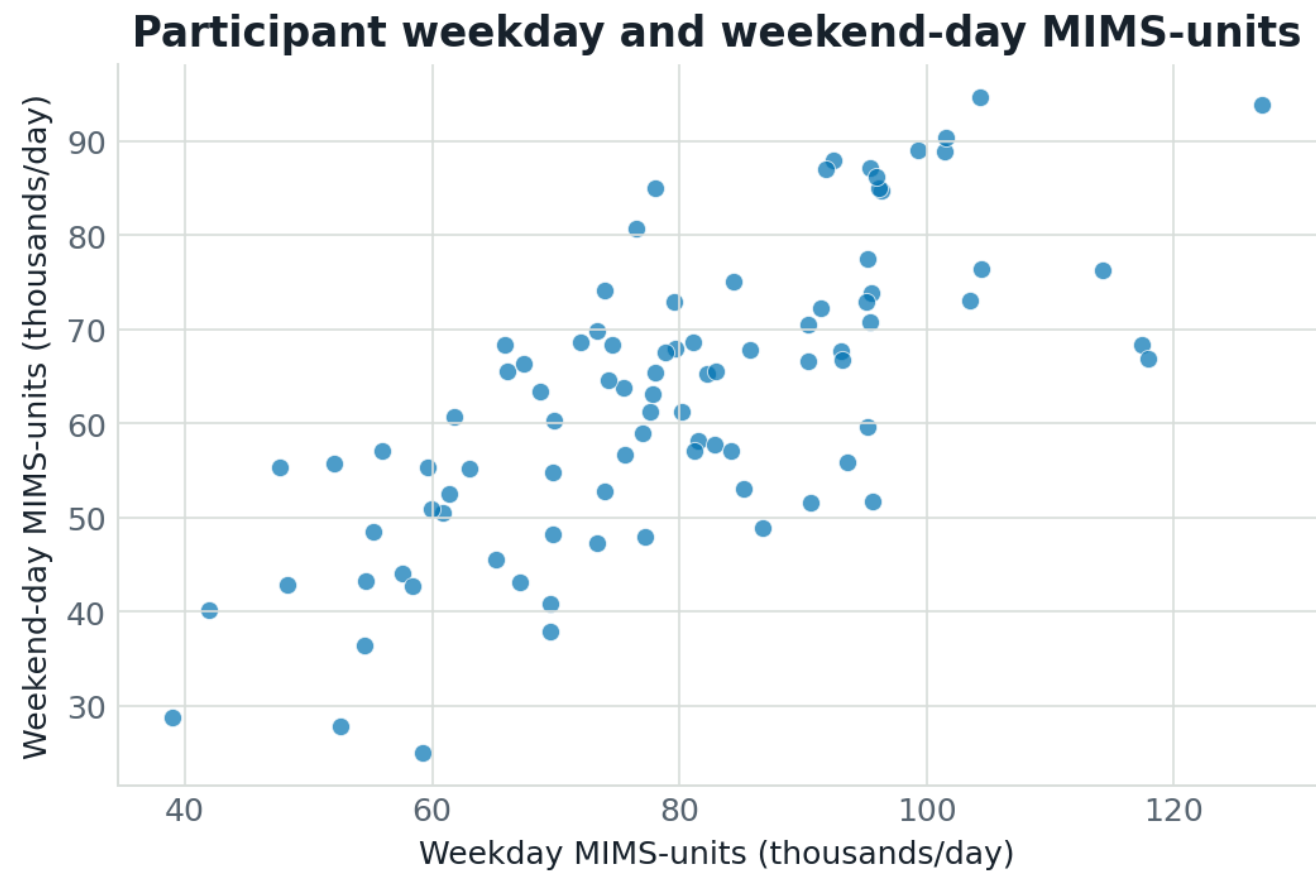
Three-part 24-hour movement-behaviour composition



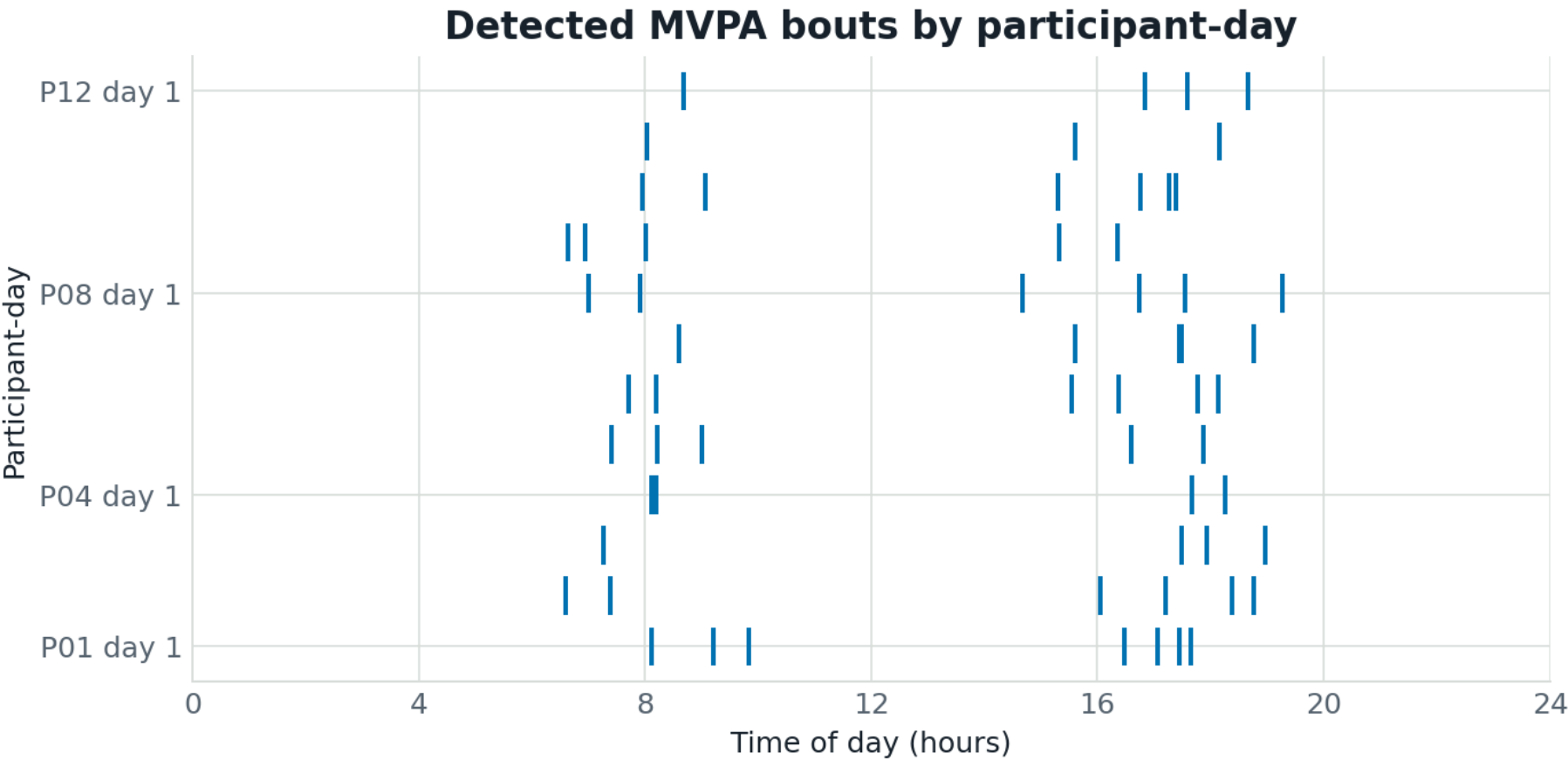
Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Ternary plot.
 Source: Simulated mock data.

Simulated example 20. Scatter plot



Simulated example 21. Event timeline or raster plot

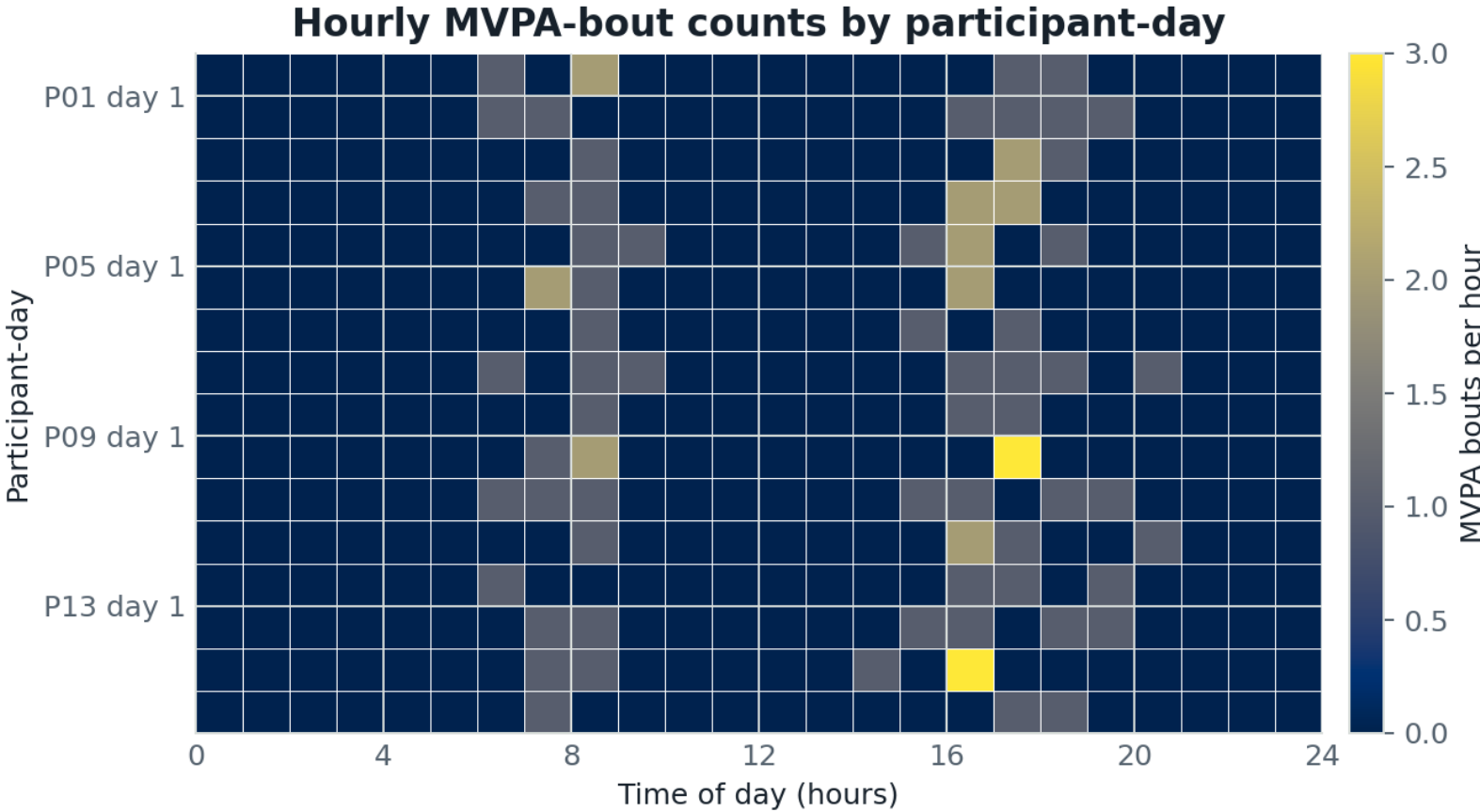


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Event timeline or raster plot.

Source: Simulated mock data.

Simulated example 22. Event raster or time-bin heatmap

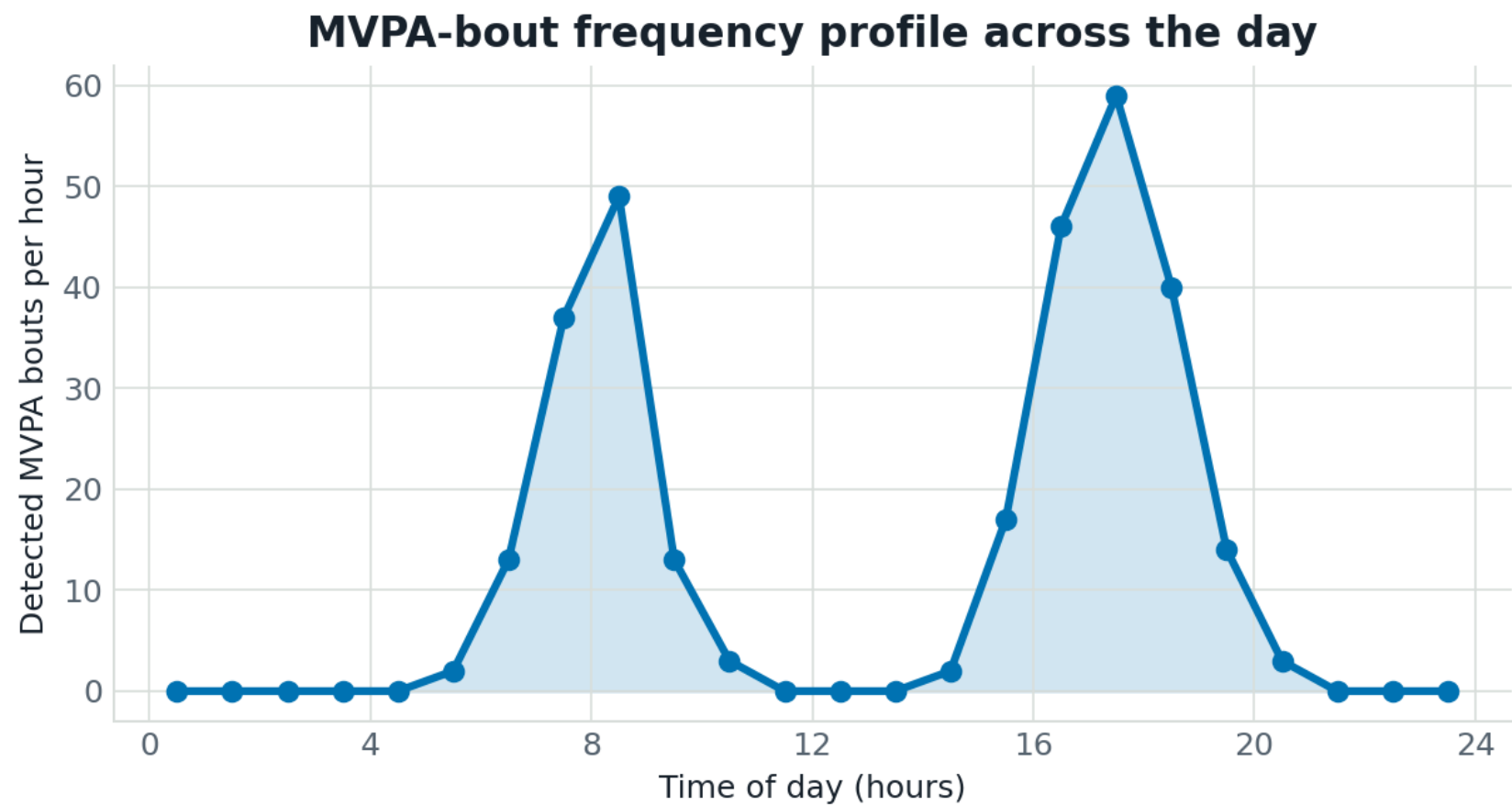


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Event raster or time-bin heatmap.

Source: Simulated mock data.

Simulated example 23. Event-frequency time profile



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Event-frequency time profile.
Source: Simulated mock data.